Parametric Equations

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Eliminating the parameter

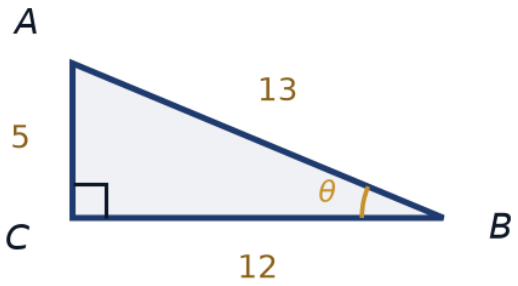
Consider the curve $x = 2t + 1$, $y = t^2$.

a Eliminate the parameter to write y as a function of x.

b Identify the curve.

2 [Foundational] • Target: Eliminating the parameter

Eliminate the parameter for $x = 3\cos t$, $y = 3\sin t$, $0 \le t < 2\pi$, and describe the curve.



3 [Foundational] • Target: Eliminating the parameter

For the line $x = t - 2$, $y = 2t + 3$:

a Find the points corresponding to $t = 0$ and $t = 2$.

b Eliminate the parameter to write the Cartesian equation.

4 [Foundational] • Target: Projectile motion

A projectile follows $x = 40t$, $y = -5t^2 + 30t$ (metres, seconds).

a When does the projectile land ($y = 0$, $t > 0$)?

b Find the horizontal range.

c Find the maximum height.

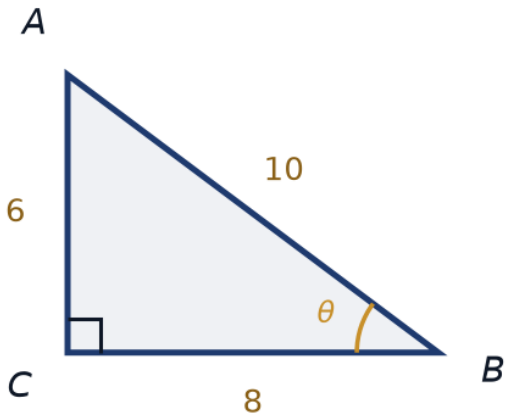
5 [Foundational] • Target: Projectile motion

A second projectile is launched with $x = 25t$, $y = -5t^2 + 40t$. Find its time of flight and maximum height.

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6 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = 4\cos t$, $y = 2\sin t$, and identify the resulting curve.



7 [Foundational] • Target: Ellipses and other parametric curves

Eliminate the parameter for $x = t^2 - 1$, $y = t^3$ by expressing t in terms of x where possible, or by finding a relation between x and y directly.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = t^2$, $y = 4t$, find $\frac{dy}{dx}$ in terms of t using $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$.

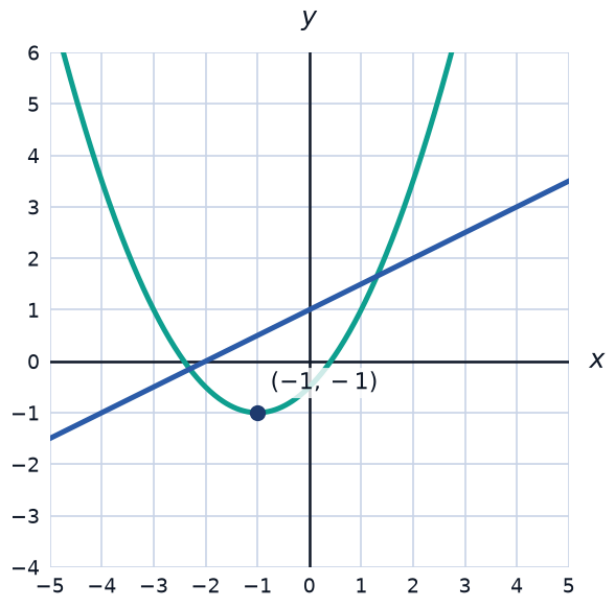
9 [Intermediate] • Target: Finding the parametric derivative

For the curve $x = \cos t$, $y = \sin t$, find the slope of the tangent line at $t = \frac{\pi}{4}$.

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10 [Intermediate] • Target: More parametric practice

A Ferris wheel of radius 10 m has its center 12 m above the ground and completes one revolution every 40 seconds. Write parametric equations for the height y of a rider above the ground as a function of time t (seconds), assuming the rider starts at the bottom of the wheel.



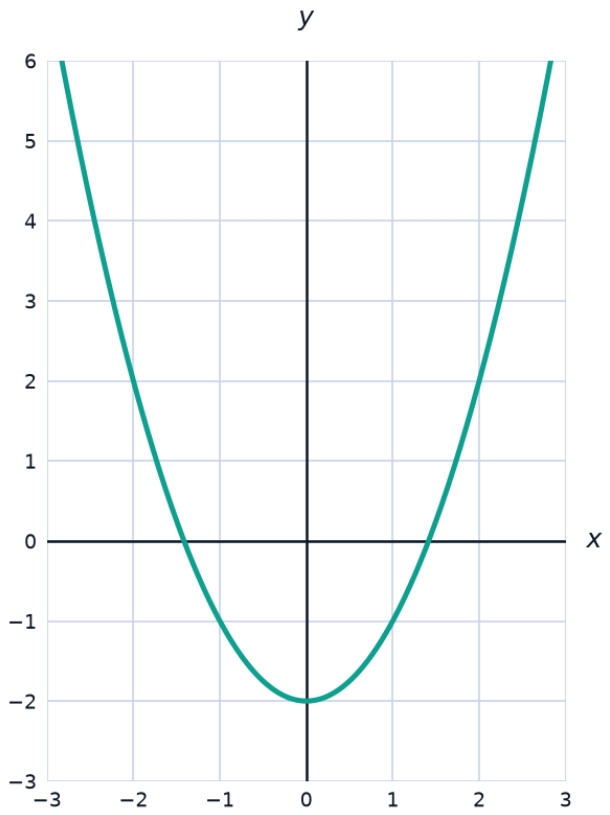
11 [Intermediate] • Target: More parametric practice

Two particles travel along paths $x_1 = t,\ y_1 = t^2$ and $x_2 = 4 - t,\ y_2 = 3t - 2$, sharing the same parameter t (time). Find the value of t at which they collide, and the point of collision.

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12 [Intermediate] • Target: Graphing a parametric curve after elimination

For $x = t$, $y = t^2 - 2$ (with t ranging over all reals), the eliminated Cartesian form $y = x^2 - 2$ is shown. Use the graph to describe the direction the curve is traced as t increases.



13 [Intermediate] • Target: Restricting the parameter domain

The curve $x = \cos t$, $y = \sin t$ for $t \in [0, \pi]$ traces only part of the unit circle. Describe which part, and state the domain and range restrictions this implies on x and y.

14 [Intermediate] • Target: Parametrizing a line segment

Write parametric equations for the line segment from $A(2, -1)$ to $B(6, 5)$, using $t \in [0, 1]$ (so $t = 0$ gives A and $t = 1$ gives B).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

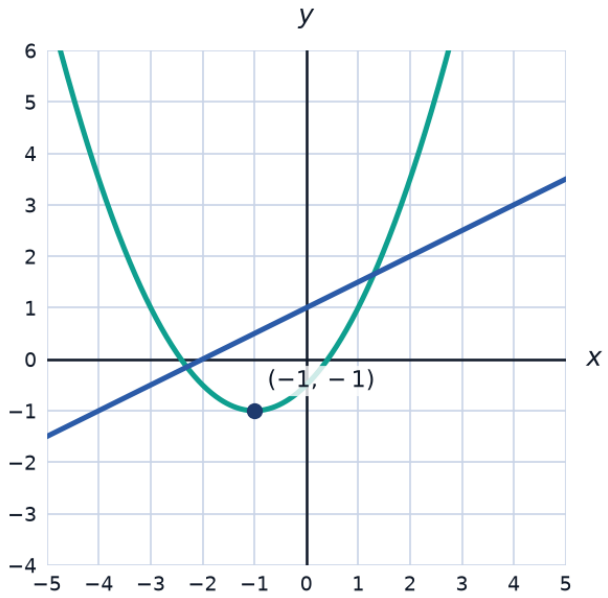
15 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 1.03 Average Rate of Change

Using the parametrization above, find the point at $t = 0.25$ (one-quarter of the way from A to B).

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16 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 2.01 Right-Triangle Trigonometry

A particle's position is $x(t) = 3t$, $y(t) = 4t$ for $t \geq 0$ (measured in meters, seconds). Find the particle's speed, and confirm it matches $\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$.



17 [Challenging] • Target: Finding the parametric derivative • Cross-Lesson: 2.02 Radian Measure and Arc Length

A population of bacteria, initially P_0 cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where t is in hours, L is the carrying capacity, and A and k are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of e^{2k} in terms of L and P_0.

18 [Challenging] • Target: More parametric practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

A radioactive isotope has a decay constant $k = 0.035\ \text{year}^{-1}$. (a) Find its half-life to the nearest tenth of a year. (b) If an initial sample contains $250\ \text{mg}$, how much remains after $40\ \text{years}$?

19 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

A geometric sequence has its first term $a_1 = 5$ and its common ratio $r = 2$. If the sum of the first n terms of this sequence is $S_n = 315$, find the value of n.

20 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Plotting and Direction of Motion

For the parametric curve $x = 2t + 1, y = t^2 - 4$ for $-2 \le t \le 2$, find the initial point and terminal point.

a. Initial: $(-3, 0)$, Terminal: $(5, 0)$

b. Initial: $(1, -4)$, Terminal: $(5, 0)$

c. Initial: $(-3, 0)$, Terminal: $(1, -4)$

d. Initial: $(0, -3)$, Terminal: $(0, 5)$

22 [Foundational] • Target: Plotting and Direction of Motion

How is orientation (direction of motion) shown on a parametric graph?

a. Arrows placed along the curve indicating the direction as the parameter t increases

b. By the slope of the curve

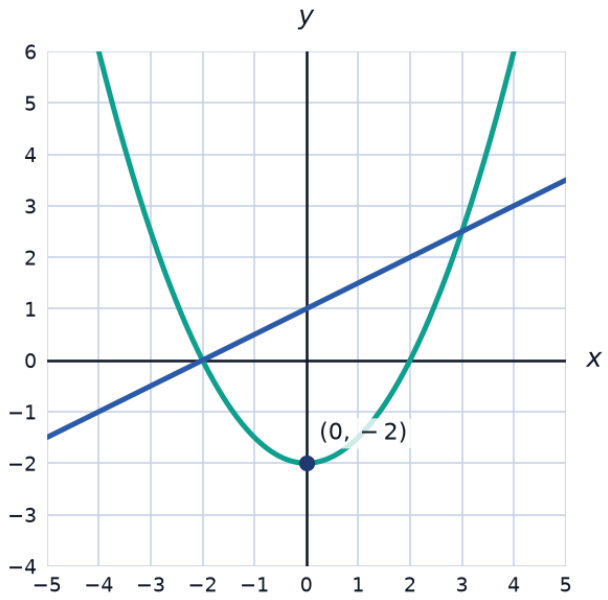
c. By the color of the line

d. By the quadrant of the initial point

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23 [Foundational] • Target: Plotting and Direction of Motion

Determine the direction of motion for $x = 3 - t, y = 2t$ as t increases.



a. Leftward and upward

b. Rightward and upward

c. Leftward and downward

d. Rightward and downward

24 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter t from $x = t - 3, y = 2t + 1$.

a. $y = 2x + 7$

b. $y = 2x - 5$

c. $y = 2x - 7$

d. $y = x + 4$

25 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter from $x = t^2, y = 3t$ for $t \geq 0$.

a. $x = \frac{y^2}{9}$ for $y \geq 0$

b. $y = 3\sqrt{x}$

c. Both of the above are correct

d. $x = 9y^2$

26 [Foundational] • Target: Eliminating the Parameter (Algebraic)

Eliminate the parameter from $x = e^t, y = 2e^{2t} - 1$.

a. $y = 2x^2 - 1$ for $x > 0$

b. $y = 2x - 1$

c. $y = e^{2x} - 1$

d. $y = 4x^2 - 1$

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27 [Foundational] • Target: Eliminating the Parameter (Trigonometric)

Eliminate the parameter from $x = 3\cos\theta, y = 3\sin\theta$.

a. $x^2 + y^2 = 9$

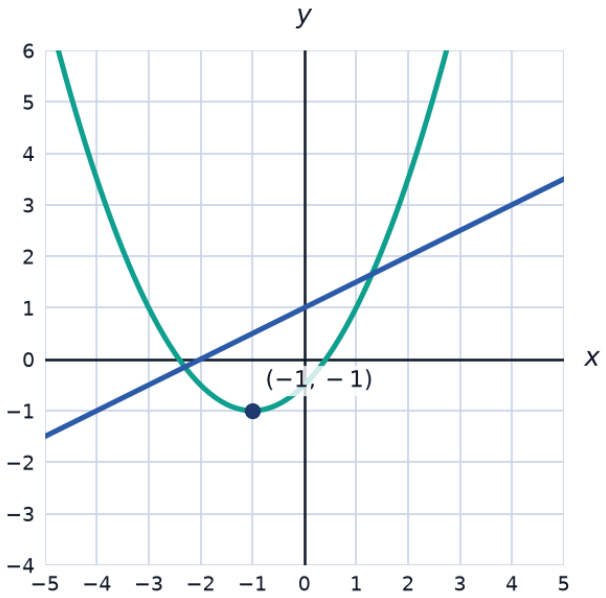
b. $x^2 + y^2 = 3$

c. $9x^2 + 9y^2 = 1$

d. $\frac{x^2}{9} - \frac{y^2}{9} = 1$

28 [Foundational] • Target: Eliminating the Parameter (Trigonometric)

Identify the Cartesian equation for $x = 4\cos\theta, y = 2\sin\theta$.



a. $\frac{x^2}{16} + \frac{y^2}{4} = 1$ (an ellipse)

b. $x^2 + y^2 = 20$

c. $\frac{x^2}{4} + \frac{y^2}{16} = 1$

d. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Eliminating the Parameter (Trigonometric)

What curve is traced by $x = 3\sec\theta, y = 2\tan\theta$?

a. A hyperbola: $\frac{x^2}{9} - \frac{y^2}{4} = 1$

b. An ellipse: $\frac{x^2}{9} + \frac{y^2}{4} = 1$

c. A parabola: $y = 2x^2$

d. A circle: $x^2 + y^2 = 13$

30 [Intermediate] • Target: Projectile Motion Applications

State the standard parametric equations for projectile motion launched with speed v_0 at angle θ from $(0, h_0)$ (with $g = 32$ ft/s^2).

a. $x(t) = (v_0\cos\theta)t, y(t) = -16t^2 + (v_0\sin\theta)t + h_0$

b. $x(t) = v_0 t, y(t) = -32t^2 + h_0$

c. $x(t) = (v_0\sin\theta)t, y(t) = -16t^2 + h_0$

d. $x(t) = (v_0\cos\theta)t, y(t) = (v_0\sin\theta)t$

31 [Intermediate] • Target: Projectile Motion Applications

A ball is kicked from ground level at 64 ft/s at an angle of 30°. At what time does it hit the ground?

a. 2 seconds

b. 1 second

c. 4 seconds

d. 3 seconds

32 [Intermediate] • Target: Projectile Motion Applications

In the previous question, what is the horizontal range of the ball?

a. $64\sqrt{3}$ ft

b. 128 ft

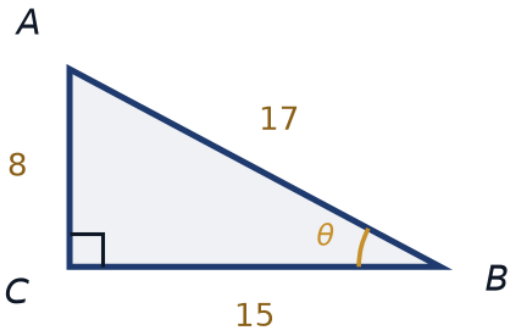
c. $32\sqrt{3}$ ft

d. 64 ft

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33 [Intermediate] • Target: Projectile Motion Applications

Eliminate the parameter: $x = 2 + 5\cos t$, $y = 0 + 5\sin t$.



a. $(x-2)^2 + (y-0)^2 = 5$

b. $(x-2)^2 + (y-0)^2 = 25$

c. $(x-0)^2 + (y-2)^2 = 25$

d. $\frac{(x-2)^2}{25} + \frac{(y-0)^2}{25} = 1$

34 [Intermediate] • Target: Eliminating the parameter

Eliminate the parameter: $x = 0 + 4\cos t$, $y = -1 + 4\sin t$.

a. $\frac{(x-0)^2}{16} + \frac{(y+1)^2}{16} = 1$

b. $(x+1)^2 + (y-0)^2 = 16$

c. $(x-0)^2 + (y+1)^2 = 4$

d. $(x-0)^2 + (y+1)^2 = 16$

35 [Intermediate] • Target: Projectile motion

Eliminate the parameter: $x = 0 + 3\cos t$, $y = -1 + 3\sin t$.

a. $(x-0)^2 + (y+1)^2 = 9$

b. $(x-0)^2 + (y+1)^2 = 3$

c. $\frac{(x-0)^2}{9} + \frac{(y+1)^2}{9} = 1$

d. $(x+1)^2 + (y-0)^2 = 9$

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36 [Intermediate] • Target: Ellipses and other parametric curves

Eliminate the parameter: $x = 2 + 2\cos t$, $y = -1 + 2\sin t$.

a. $(x-2)^2 + (y+1)^2 = 4$

b. $(x-2)^2 + (y+1)^2 = 2$

c. $\frac{(x-2)^2}{4} + \frac{(y+1)^2}{4} = 1$

d. $(x+1)^2 + (y-2)^2 = 4$

37 [Intermediate] • Target: Finding the parametric derivative

Eliminate the parameter: $x = t$, $y = -1t + 5$.

a. $x = -1y + 5$

b. $y = -1x$

c. $y = -1x + 5$

d. $y = 1x + 5$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: More parametric practice • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Two vectors are given by $\vec{u} = 3\hat{i} + k\hat{j}$ and $\vec{v} = -\hat{i} + 4\hat{j}$. If $\vec{u}$ and $\vec{v}$ are orthogonal, find the value of the scalar parameter k.

a. 3/4

b. 4/3

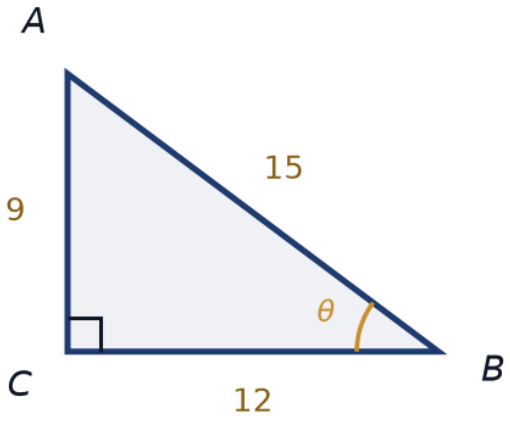
c. -3/4

d. -4/3

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39 [Challenging] • Target: Graphing a parametric curve after elimination • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$.



a. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

b. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

c. $x = \frac{\pi}{2}$ only

d. $x = 0, \pi$

40 [Challenging] • Target: Restricting the parameter domain • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 3)

a. $x = \frac{\pi}{2}$ only

b. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

c. $x = 0, \pi$

d. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

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41 [Challenging] • Target: Parametrizing a line segment • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 6)

a. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

b. $x = \frac{\pi}{2}$ only

c. $x = 0, \pi$

d. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

42 [Challenging] • Target: Parametric speed and motion review • Cross-Lesson: 3.03 Sum and Difference Formulas

Solve $\cos(2x) + \sin x = 0$ on $[0, 2\pi)$. (attempt 7)

a. $x = 0, \pi$

b. $x = \frac{7\pi}{6}, \frac{11\pi}{6}$ only

c. $x = \frac{\pi}{2}$ only

d. $x = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$

43 [Challenging] • Target: Plotting and Direction of Motion • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

Find the projection of $\mathbf{u} = \langle 6, -2 \rangle$ onto $\mathbf{v} = \langle 2, -1 \rangle$.

a. $\left\langle \frac{33}{5}, -\frac{14}{5} \right\rangle$

b. $\left\langle -\frac{14}{5}, \frac{28}{5} \right\rangle$

c. $\frac{14}{5}$

d. $\left\langle \frac{28}{5}, -\frac{14}{5} \right\rangle$

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44 [Challenging] • Target: Eliminating the Parameter (Algebraic) • Cross-Lesson: 4.01 Circles and Parabolas

Find the projection of $\mathbf{u} = \langle 3, 0 \rangle$ onto $\mathbf{v} = \langle 1, -1 \rangle$.

a. $\left\langle \frac{5}{2}, -\frac{3}{2} \right\rangle$

b. $\frac{3}{2}$

c. $\left\langle \frac{3}{2}, -\frac{3}{2} \right\rangle$

d. $\left\langle -\frac{3}{2}, \frac{3}{2} \right\rangle$

45 [Challenging] • Target: Eliminating the Parameter (Trigonometric) • Cross-Lesson: 4.02 Ellipses

Find the projection of $\mathbf{u} = \langle 3, 2 \rangle$ onto $\mathbf{v} = \langle 2, 2 \rangle$.

a. $\left\langle \frac{3}{2}, \frac{5}{2} \right\rangle$

b. $\left\langle \frac{5}{2}, \frac{5}{2} \right\rangle$

c. $\left\langle \frac{7}{2}, \frac{5}{2} \right\rangle$

d. $\frac{5}{4}$

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